A

Project Report

On

[Stock Prediction & Analysis]

Submitted in partial fulfillment of the course

Supervised and Unsupervised Learning [24CAI0203]

of

Computer Science & Engineering (AI&ML)

during January-June, 2026

Submitted by:

2410992916(Vishesh)

2410992597 (Bhrat Arora)

CHITKARA UNIVERSITY

CHANDIGARH-PATIALA NATIONAL HIGHWAY
RAJPURA (PATIALA) PUNJAB-140401 (INDIA)

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DECLARATION OF PROJECT

We, the undersigned students of the Department of Computer Science & Engineering – AI&ML at Chitkara University, Punjab, hereby declare that we have successfully completed the project titled "**Stock Prediction & Analysis"** as part of the requirements for the Course **Supervised and Unsupervised Learning [24CAI0203]**.

We confirm that the project was conducted during the period [23-03-2026] to [01-05-2026] and has been prepared in partial fulfillment of the course *Supervised and Unsupervised Learning* for submission to the Department of Computer Science and Engineering (Artificial Intelligence).

We affirm that the work submitted is our original effort and has not been copied or reproduced from any other source, except where due acknowledgment has been made. We also confirm that all the resources, references, and materials used have been properly cited in the project.

We undertake to abide by the rules and regulations of Chitkara University, Punjab and accept full responsibility for the authenticity and validity of the project submitted.

Date: [28-04-2026]

Signed by:

2410992916, Vishesh (Team Member 1)
2410992597, Bhrat Arora (Team Member 2)

Abstract

Stock market prediction is one of the most challenging problems in financial analytics due to the highly dynamic and non-linear nature of financial markets. Investors and analysts continuously seek accurate methods to analyze stock price trends and make informed investment decisions.

This project focuses on predicting stock price trends for **multiple companies across different sectors** using machine learning techniques. Historical stock market data was collected from **Yahoo Finance** using the Python library **yfinance**.

The dataset includes several companies from sectors such as **banking, IT, automobile, pharmaceutical, and FMCG**. Example stocks analyzed include **TCS, Infosys, Wipro, HDFC Bank, ICICI Bank, SBI, Maruti Suzuki, Tata Motors, M&M, Nestle India, Sun Pharma, Dr. Reddy, and Cipla**.

Various preprocessing techniques such as **data cleaning, feature engineering, normalization, and feature scaling** were applied to prepare the dataset. Important financial indicators such as **moving averages, daily returns, and trading volume trends** were generated to capture patterns in stock price movements.

Multiple machine learning algorithms including **K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Decision Tree, and Random Forest** were implemented to analyze and predict stock price behavior.

The results demonstrate that machine learning models can effectively identify patterns in historical financial data and assist in forecasting stock price trends. This project highlights the potential of machine learning in financial analysis and investment strategy development.

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Introduction

The stock market plays a crucial role in the global economy by enabling companies to raise capital and investors to grow their wealth. However, stock prices are highly volatile and influenced by many factors such as economic indicators, company performance, market sentiment, geopolitical events, and investor behavior.

Predicting stock prices accurately is extremely difficult due to the dynamic and unpredictable nature of financial markets. Traditional statistical approaches often fail to capture complex patterns present in stock market data.

Machine learning provides powerful techniques to analyze large datasets and identify hidden patterns. By training algorithms on historical data, predictive models can estimate future stock price movements.

In this project, machine learning models are applied to analyze **multiple stocks from different sectors of the Indian stock market**. Historical data for companies such as **TCS, Infosys, Wipro, HDFC Bank, ICICI Bank, SBI, Maruti Suzuki, Tata Motors, M&M, Nestle India, Sun Pharma, Dr. Reddy, and Cipla** were collected and analyzed.

The goal of this project is to build a machine learning pipeline capable of predicting stock price behavior and identifying trends across multiple companies.

Problem Statement

Stock market prediction is a complex problem due to the highly volatile nature of financial markets. Investors often rely on historical data and market indicators to guide their decisions, but manual analysis of large financial datasets is time-consuming and inefficient.

The objective of this project is to develop a **machine learning system that analyzes historical stock market data for multiple companies and predicts future stock price trends**.

The key challenges addressed in this project include:

- Handling large volumes of financial time-series data
- Extracting meaningful features from raw stock market data
- Training machine learning models for prediction
- Evaluating model performance across multiple stocks

By addressing these challenges, the system aims to provide insights into stock market behavior and assist investors in making data-driven decisions.

Methodology & Technical Details

The proposed system follows a structured machine learning pipeline to analyze historical stock market data and predict stock price trends for multiple companies. The methodology consists of several stages including data collection, preprocessing, feature engineering, model training, evaluation, and visualization**1. Data Collection**

1. Data Collection

The first step in the project is collecting historical stock market data for multiple companies. The data was obtained from Yahoo Finance using the Python library yfinance.

The dataset includes companies from different sectors of the Indian stock market such as:

Information Technology Sector

- TCS (Tata Consultancy Services)
- Infosys
- Wipro

Banking Sector

- HDFC Bank
- ICICI Bank
- State Bank of India

Automobile Sector

- Maruti Suzuki
- Tata Motors
- Mahindra & Mahindra

Pharmaceutical Sector

- Sun Pharma
- Dr. Reddy's Laboratories
- Cipla

FMCG Sector

- Nestle India

The historical stock data ranges from 2015 to 2026 and includes important attributes such as:

- Date
- Open Price
- High Price
- Low Price
- Close Price
- Adjusted Close Price

- Volume

This data forms the foundation for the machine learning models used in this project.

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2. Data Preprocessing

Raw financial data often contains inconsistencies, missing values, or irrelevant information. Therefore, preprocessing is required to prepare the dataset for analysis.

The preprocessing steps performed in this project include:

- Handling missing values in the dataset
- Removing duplicate records
- Converting date columns into proper datetime format
- Sorting the dataset chronologically
- Ensuring numerical columns are correctly formatted

These preprocessing steps ensure that the dataset is clean and ready for further analysis.

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3. Feature Engineering

Feature engineering is an important step that helps improve the predictive capability of machine learning models.

In this project, additional features were created from the raw stock data, including:

- Daily Returns – percentage change between closing prices
- Moving Averages (7-day, 30-day, and 90-day averages)
- Price Change Percentage
- Volume Trends

These features help the model capture important patterns and trends in stock price movements.

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4. Model Development

Several machine learning algorithms were implemented to predict stock price behavior. These models were trained using the processed dataset.

The models used in this project include:

K-Nearest Neighbors (KNN)

KNN predicts stock price trends based on similarity between historical data points.

Support Vector Machine (SVM)

SVM identifies optimal decision boundaries that separate different classes in the dataset.

Decision Tree

Decision Tree models split the dataset into different branches based on feature conditions.

Random Forest
Random Forest is an ensemble learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Each model was trained and tested using the prepared dataset to evaluate its performance.

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5. Model Evaluation

After training the machine learning models, their performance was evaluated using standard evaluation metrics.

The evaluation metrics used include:

- Mean Squared Error (MSE) – measures the average squared difference between predicted and actual values.
- Mean Absolute Error (MAE) – measures the average absolute difference between predicted and actual values.
- Accuracy Score – used for classification-based predictions.

These metrics help determine how well the models perform in predicting stock price trends.

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6. Visualization & Insights

To better understand stock price patterns and model predictions, several visualizations were generated using Python libraries such as Matplotlib and Seaborn.

The visualizations include:

- Stock price trend graphs
- Moving average comparisons
- Prediction vs actual price charts
- Volume trend analysis

These graphs help interpret the results and provide insights into stock market behavior.

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7. System Workflow

The overall workflow of the system can be summarized as follows:

Data Collection → Data Preprocessing → Feature Engineering → Model Training → Model Evaluation → Visualization → Insights Generation

This pipeline enables efficient analysis of stock market data and accurate prediction of stock price trends using machine learning techniques.

Dataset Description & Preparation

1. Dataset Description

The dataset consists of historical stock market data collected from Yahoo Finance.

Dataset Attributes

Each dataset contains the following columns:

• Date
• Open Price
• High Price
• Low Price
• Close Price
• Adjusted Close Price
• Volume

Data Preparation Steps

1. Data Cleaning

• Removing missing values
• Fixing incorrect data types

2. Feature Creation

• Moving averages
• Daily returns

3. **Normalization**

• Scaling features to improve model performance

4. **Train-Test Split**

The dataset was divided into training and testing sets.

Project Advantages & Limitation of Models

Advantages

• Provides data-driven analysis of stock market trends

• Uses multiple machine learning algorithms

• Supports analysis of multiple companies across sectors

• Can identify patterns in financial time-series data

• Scalable to additional stocks

Limitations

• Stock markets are inherently unpredictable

• Models rely only on historical data

• External factors such as news and economic events are not included

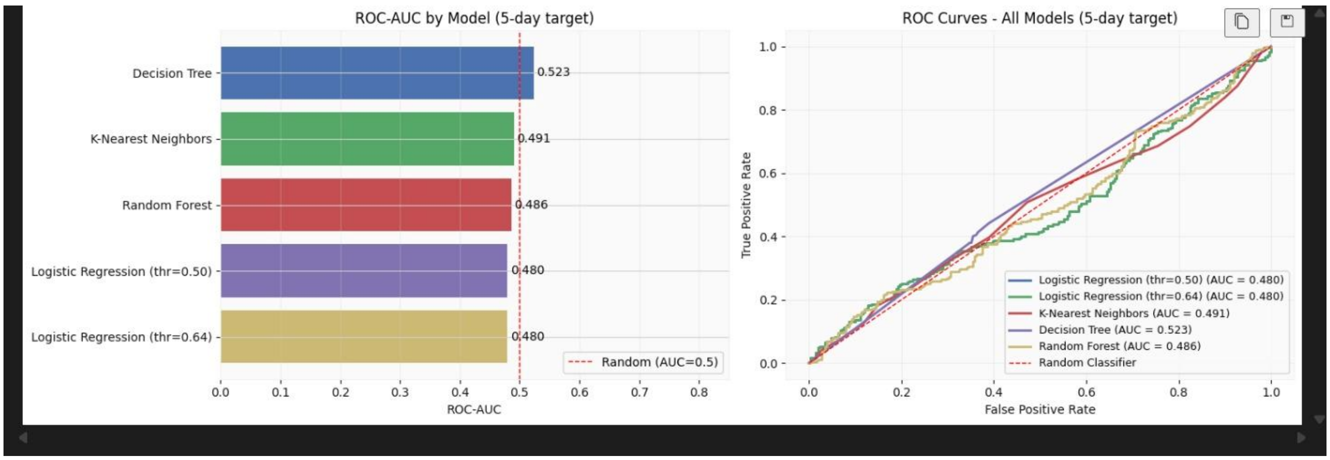
• Prediction accuracy may vary depending on market volatility

Results & Discussions

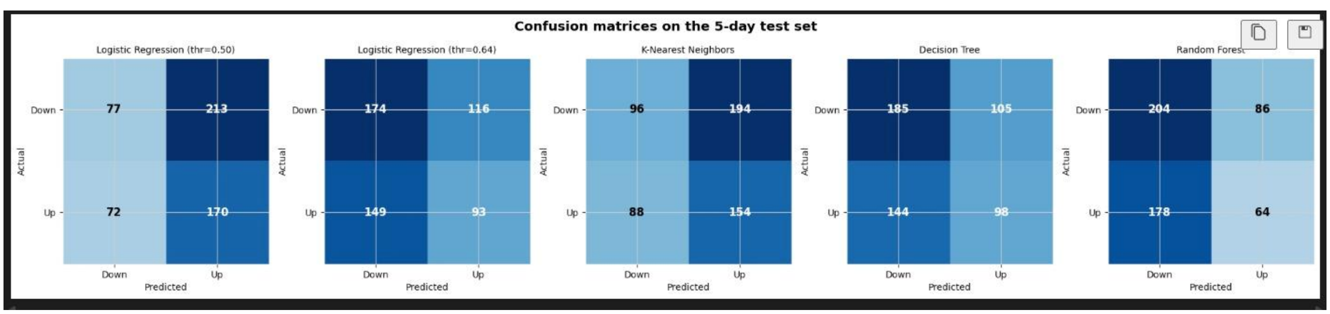
The machine learning models were trained using historical stock data from multiple companies.

Model Observations

• **Random Forest** provided the best prediction performance.
• **Decision Trees** captured non-linear relationships effectively.
• **KNN** worked well for local pattern recognition.
• **SVM** provided stable decision boundaries.



Confusion Matrix Analysis



Confusion matrices for Logistic Regression, K-Nearest Neighbors, Decision Tree, and Random Forest models on the test dataset.

The confusion matrix provides a detailed view of the classification performance of each machine learning model. It shows the number of correct and incorrect predictions for both upward and downward stock movements.

The matrix consists of:

- **True Positives (TP):** Correct prediction of upward stock movement

- **True Negatives (TN):** Correct prediction of downward movement
- **False Positives (FP):** Incorrect prediction of upward movement
- **False Negatives (FN):** Incorrect prediction of downward movement

By analyzing the confusion matrix, we can understand how each model performs in predicting stock trends and where misclassifications occur.

Key Insights

• Moving averages significantly improved prediction accuracy.
• Trading volume influenced stock price movement.
• Ensemble models such as Random Forest performed better than individual models.

Conclusion with Future Scope

Conclusion

This project successfully developed a machine learning pipeline to analyze and predict stock price trends for **multiple companies across different sectors**.

The results demonstrate that machine learning algorithms can effectively analyze financial time-series data and identify patterns in stock price movements.

This system can help investors and analysts gain insights into market trends and support data-driven investment decisions.

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...   === 5-Day Target: Final Test-Set Comparison ===
                          Model  Accuracy  ROC-AUC  Precision_1  Recall_1   F1_1
                  Decision Tree    0.5320   0.5235       0.4828    0.4050 0.4404
            K-Nearest Neighbors    0.4699   0.4909       0.4425    0.6364 0.5220
                  Random Forest    0.5038   0.4859       0.4267    0.2645 0.3265
   Logistic Regression (thr=0.50)    0.4643   0.4796       0.4439    0.7025 0.5440
   Logistic Regression (thr=0.64)    0.5019   0.4796       0.4450    0.3843 0.4124
```

Future Scope

• Implementing **Deep Learning models such as LSTM** for time-series prediction

• Integrating **financial news sentiment analysis**

• Developing a **real-time stock prediction dashboard**

• Expanding the dataset to include **global stock markets**

References

1. Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow
2. Scikit-Learn Documentation
3. Pandas Documentation
4. NumPy Documentation
5. Yahoo Finance API (yfinance)